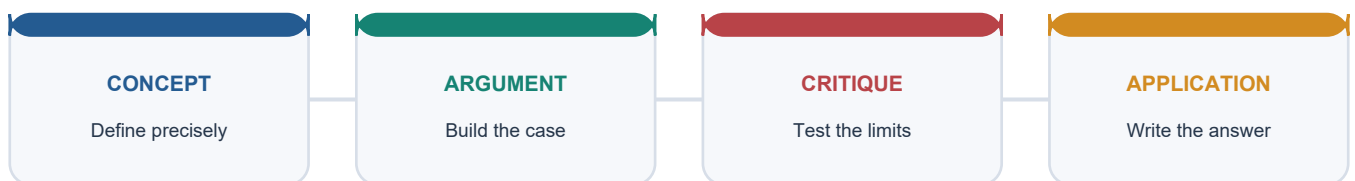


SOLVED PRACTICE WORKBOOK

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook

Economy | UPSC Prelims and GS-III

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook 4

BASIC MCQS / REMEDIATION 4

MCQ 1	4
MCQ 2	4
MCQ 3	4
MCQ 4	5
MCQ 5	5
MCQ 6	5
MCQ 7	6
MCQ 8	6
MCQ 9	6
MCQ 10	7
MCQ 11	7
MCQ 12	7
MCQ 13	8
MCQ 14	8
MCQ 15	8
MCQ 16	9
MCQ 17	9
MCQ 18	9
MCQ 19	10
MCQ 20	10
MCQ 21	10
MCQ 22	11
MCQ 23	11
MCQ 24	11
MCQ 25	12
MCQ 26	12
MCQ 27	12

MCQ 28	13
MCQ 29	13
MCQ 30	13
MCQ 31	14
MCQ 32	14
PYQS AND ANSWER PRACTICE	14
VERIFIED ROUTED MAINS DEMANDS	14
2018 GS-III ROUTE - SIKKIM ORGANIC TRANSITION	14
2019 GS-III ROUTE - INTEGRATED FARMING SYSTEMS	15
2019 GS-III ROUTE - WATERSHED DEVELOPMENT	15
2020 GS-III ROUTE - JAL SHAKTI ABHIYAN	15
2020 GS-III ROUTE - WATER STORAGE AND IRRIGATION	15
2021 GS-III ROUTE - MICRO-IRRIGATION	15
2022 GS-III ROUTE - IFS FOR SMALL FARMERS	16
2024 GS-III ROUTE - IRRIGATION CHALLENGES	16
2025 GS-III ROUTE - GROUNDWATER DEPLETION	16
OBJECTIVE PYQ ROUTING - KEY WITHHELD WHERE NOT OFFICIALLY VERIFIED	16
ORIGINAL MAINS PRACTICE WITH MODEL SOLUTIONS	17
ORIGINAL MAINS 1 - 10 MARKS	17
ORIGINAL MAINS 2 - 10 MARKS	17
ORIGINAL MAINS 3 - 15 MARKS	17
ORIGINAL MAINS 4 - 15 MARKS	18
ORIGINAL MAINS 5 - 20 MARKS	18
ORIGINAL MAINS 6 - 20 MARKS	19

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook

Exactly 32 original questions appear before PYQs. Correct answers rotate A -> B -> C -> D eight times.

BASIC MCQS / REMEDIATION

MCQ 1

Which statement correctly distinguishes irrigation potential and utilisation?

A. Potential is designed capacity; utilisation is the area actually served under the system. B. Potential is crop output per litre. C. Utilisation is identical to gross irrigated area in every dataset. D. Potential proves equitable tail-end delivery.

Answer: A.

Option-wise explanation:

- **A - Correct:** It separates infrastructure capacity from realised service. The option states: Potential is designed capacity; utilisation is the area actually served under the system.
- **B - Incorrect:** That is closer to water productivity. The option states: Potential is crop output per litre.
- **C - Incorrect:** Statistical concepts have distinct denominators. The option states: Utilisation is identical to gross irrigated area in every dataset.
- **D - Incorrect:** Distribution still needs evidence. The option states: Potential proves equitable tail-end delivery.

Examiner trap 1: Created capacity is not reliable delivery.

MCQ 2

Irrigation intensity is calculated as:

A. Net irrigated area divided by gross sown area. B. Gross irrigated area divided by net irrigated area multiplied by 100. C. Water withdrawn divided by crop output. D. Potential utilised divided by potential created.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** That uses an unrelated denominator. The option states: Net irrigated area divided by gross sown area.
- **B - Correct:** It counts repeated irrigated crop use on irrigated land. The option states: Gross irrigated area divided by net irrigated area multiplied by 100.
- **C - Incorrect:** This is inverse water productivity. The option states: Water withdrawn divided by crop output.
- **D - Incorrect:** That is a project-utilisation ratio. The option states: Potential utilised divided by potential created.

Examiner trap 2: Intensity is not efficiency.

MCQ 3

Which is a distinctive risk of canal irrigation?

A. Universal elimination of groundwater pumping. B. No head-tail distribution problem. C. Waterlogging and salinity where excess application and poor drainage raise the water table. D. Absence of any recharge benefit from seepage.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Canals do not eliminate every source. The option states: Universal elimination of groundwater pumping.
- **B - Incorrect:** Head-tail inequity is a standard concern. The option states: No head-tail distribution problem.

- **C - Correct:** Canal command areas need drainage and distribution management. The option states: Waterlogging and salinity where excess application and poor drainage raise the water table.
- **D - Incorrect:** Seepage can sometimes recharge aquifers. The option states: Absence of any recharge benefit from seepage.

Examiner trap 3: Canal expansion can create land-quality externalities.

MCQ 4

A watershed is best defined as:

A. The area legally notified under an APMC Act. B. Only land irrigated by one canal outlet. C. A bank's agricultural lending district. D. A geo-hydrological area draining runoff to a common outlet.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** That is a market-law concept. The option states: The area legally notified under an APMC Act.
- **B - Incorrect:** A command outlet and watershed differ. The option states: Only land irrigated by one canal outlet.
- **C - Incorrect:** Credit geography is unrelated. The option states: A bank's agricultural lending district.
- **D - Correct:** The common drainage unit justifies ridge-to-valley treatment. The option states: A geo-hydrological area draining runoff to a common outlet.

Examiner trap 4: Watershed and command area are not synonyms.

MCQ 5

Which pair is correctly matched?

A. Drip-frequent controlled application near the root zone. B. Sprinkler-water delivered only through underground root emitters. C. Canal-pressurised micro-irrigation on every farm. D. Tensiometer-device that pumps groundwater.

Answer: A.

Option-wise explanation:

- **A - Correct:** Drip uses emitters for precise application. The option states: Drip-frequent controlled application near the root zone.
- **B - Incorrect:** That describes drip rather than all sprinklers. The option states: Sprinkler-water delivered only through underground root emitters.
- **C - Incorrect:** Canals are conveyance systems. The option states: Canal-pressurised micro-irrigation on every farm.
- **D - Incorrect:** A tensiometer measures soil-water tension. The option states: Tensiometer-device that pumps groundwater.

Examiner trap 5: Micro-irrigation contains distinct methods.

MCQ 6

Why can micro-irrigation fail to reduce basin extraction?

A. Drip necessarily increases evaporation. B. Lower water use per hectare may induce expansion of irrigated area or water-intensive crops. C. Sprinklers cannot be metered. D. Micro-irrigation prohibits fertigation.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** Drip often reduces non-beneficial loss. The option states: Drip necessarily increases evaporation.
- **B - Correct:** The rebound effect separates field efficiency from total withdrawal. The option states: Lower water use per hectare may induce expansion of irrigated area or water-intensive crops.
- **C - Incorrect:** Metering is possible. The option states: Sprinklers cannot be metered.
- **D - Incorrect:** Fertigation is a common capability. The option states: Micro-irrigation prohibits fertigation.

Examiner trap 6: Per-hectare saving is not basin saving.

MCQ 7

Which statement about Per Drop More Crop is current?

A. It is the major-dam component of PMKSY. B. It was repealed with the 2020 farm laws. C. It supports micro-irrigation and has been implemented under RKVY from 2022-23. D. It is a crop-insurance premium subsidy.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** AIBP addresses major and medium projects. The option states: It is the major-dam component of PMKSY.
- **B - Incorrect:** The farm-law repeal is unrelated. The option states: It was repealed with the 2020 farm laws.
- **C - Correct:** The policy objective continues under a changed administrative umbrella. The option states: It supports micro-irrigation and has been implemented under RKVY from 2022-23.
- **D - Incorrect:** PMFBY is the insurance scheme. The option states: It is a crop-insurance premium subsidy.

Examiner trap 7: Do not freeze the older PMKSY placement.

MCQ 8

Participatory irrigation management primarily uses:

A. CCI to set canal rotations. B. WDRA to insure crops. C. FPOs to regulate river basins by company law. D. Water Users Associations for local distribution, maintenance and accountability roles.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** CCI has an antitrust mandate. The option states: CCI to set canal rotations.
- **B - Incorrect:** WDRA regulates warehouses. The option states: WDRA to insure crops.
- **C - Incorrect:** FPOs have a producer-business purpose. The option states: FPOs to regulate river basins by company law.
- **D - Correct:** User institutions can coordinate a shared command service. The option states: Water Users Associations for local distribution, maintenance and accountability roles.

Examiner trap 8: Creating a WUA does not prove effective participation.

MCQ 9

Which seed-class sequence is correct?

A. Nucleus, breeder, foundation and certified seed. B. Certified, nucleus, foundation and breeder. C. Foundation, certified, nucleus and breeder. D. Breeder, certified, nucleus and foundation.

Answer: A.

Option-wise explanation:

- **A - Correct:** The sequence expands genetically identified material in controlled stages. The option states: Nucleus, breeder, foundation and certified seed.
- **B - Incorrect:** It reverses and scrambles pedigree. The option states: Certified, nucleus, foundation and breeder.
- **C - Incorrect:** Certified seed does not precede nucleus. The option states: Foundation, certified, nucleus and breeder.
- **D - Incorrect:** Foundation normally precedes certified. The option states: Breeder, certified, nucleus and foundation.

Examiner trap 9: Seed multiplication has a traceable pedigree.

MCQ 10

What is the current legal status of the Seeds Bill, 2025?

A. It became the Seeds Act automatically on publication. B. It was circulated as a draft for consultation and had not replaced the Seeds Act, 1966 by the cutoff. C. It repealed the Seeds Rules, 1968 in 2019. D. It is a State model law like APLM.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** Consultation is not enactment. The option states: It became the Seeds Act automatically on publication.
- **B - Correct:** Draft and enacted law are separate statuses. The option states: It was circulated as a draft for consultation and had not replaced the Seeds Act, 1966 by the cutoff.
- **C - Incorrect:** The chronology is incorrect. The option states: It repealed the Seeds Rules, 1968 in 2019.
- **D - Incorrect:** Seed regulation is not that marketing model. The option states: It is a State model law like APLM.

Examiner trap 10: The Seeds Act, 1966 remains the legal anchor.

MCQ 11

Which statement correctly describes formal and informal seed systems?

A. Informal seed is always illegal. B. Formal seed guarantees suitability in every agro-climate. C. Formal channels use organised quality control and labels; informal channels include farm saving and local exchange. D. Farm-saved seed is necessarily certified.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Informality is not automatically illegality. The option states: Informal seed is always illegal.
- **B - Incorrect:** Suitability remains location-specific. The option states: Formal seed guarantees suitability in every agro-climate.
- **C - Correct:** The distinction concerns channels and assurance, not a blanket quality verdict. The option states: Formal channels use organised quality control and labels; informal channels include farm saving and local exchange.
- **D - Incorrect:** Certification requires prescribed processes. The option states: Farm-saved seed is necessarily certified.

Examiner trap 11: Access, diversity and verification must be balanced.

MCQ 12

Which nutrient match is correct?

A. Nitrogen supplies all N, P and K. B. Phosphorus is only a pesticide. C. Potassium is the primary source of plant protein nitrogen. D. Potassium supports water regulation, stress tolerance and quality functions.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** No single nutrient supplies all three. The option states: Nitrogen supplies all N, P and K.
- **B - Incorrect:** Phosphorus is a nutrient. The option states: Phosphorus is only a pesticide.
- **C - Incorrect:** Nitrogen is the key protein nutrient. The option states: Potassium is the primary source of plant protein nitrogen.
- **D - Correct:** K has distinct osmotic and stress roles. The option states: Potassium supports water regulation, stress tolerance and quality functions.

Examiner trap 12: N, P and K are not interchangeable.

MCQ 13

Which statement about fertiliser subsidy is correct?

A. Urea has controlled MRP under a separate subsidy regime; eligible P&K fertilisers receive NBS. B. Urea is included in NBS exactly like DAP. C. All P&K MRPs are statutorily fixed like urea. D. NBS rates never change by season.

Answer: A.

Option-wise explanation:

- **A - Correct:** India uses different price and subsidy architectures. The option states: Urea has controlled MRP under a separate subsidy regime; eligible P&K fertilisers receive NBS.
- **B - Incorrect:** Urea remains outside NBS. The option states: Urea is included in NBS exactly like DAP.
- **C - Incorrect:** P&K MRP is decontrolled subject to policy oversight. The option states: All P&K MRPs are statutorily fixed like urea.
- **D - Incorrect:** NBS rates are notified periodically. The option states: NBS rates never change by season.

Examiner trap 13: Relative price regimes affect nutrient balance.

MCQ 14

Fertiliser DBT means:

A. Every farmer receives the subsidy in a bank account before purchase. B. Government reimburses companies after authenticated retail sale through PoS; it is not cash paid to the farmer. C. Dealer subsidy is unrelated to actual sale. D. Only imported fertiliser can use PoS.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** The farmer receives a subsidised price. The option states: Every farmer receives the subsidy in a bank account before purchase.
- **B - Correct:** The transaction-triggered company payment is the defining feature. The option states: Government reimburses companies after authenticated retail sale through PoS; it is not cash paid to the farmer.
- **C - Incorrect:** Retail sale validation is central. The option states: Dealer subsidy is unrelated to actual sale.
- **D - Incorrect:** The system covers eligible domestic and imported supplies. The option states: Only imported fertiliser can use PoS.

Examiner trap 14: Do not confuse DBT labels across schemes.

MCQ 15

A Soil Health Card should be used as:

A. A permanent guarantee of soil fertility. B. A title deed for agricultural land. C. A dated soil-test report guiding crop- and site-specific nutrient decisions. D. A crop-insurance claim certificate.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Soils change with management. The option states: A permanent guarantee of soil fertility.
- **B - Incorrect:** Land title has another record system. The option states: A title deed for agricultural land.
- **C - Correct:** Sampling date and laboratory result determine relevance. The option states: A dated soil-test report guiding crop- and site-specific nutrient decisions.
- **D - Incorrect:** Insurance uses separate evidence. The option states: A crop-insurance claim certificate.

Examiner trap 15: One sample cannot diagnose every plot forever.

MCQ 16

Which statement about nano urea is evidence-disciplined?

A. One bottle universally replaces one bag in every crop. B. Regulatory approval proves identical yield in every soil. C. Nano urea supplies phosphorus and potassium automatically. D. Use should follow crop-soil protocols; notification does not prove universal replacement of conventional urea.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** Universal equivalence is unsupported. The option states: One bottle universally replaces one bag in every crop.
- **B - Incorrect:** Approval is not universal efficacy. The option states: Regulatory approval proves identical yield in every soil.
- **C - Incorrect:** Nano urea is a nitrogen product. The option states: Nano urea supplies phosphorus and potassium automatically.
- **D - Correct:** Official and ICAR evidence requires context and rejects blanket replacement. The option states: Use should follow crop-soil protocols; notification does not prove universal replacement of conventional urea.

Examiner trap 16: Product launch is not agronomic proof.

MCQ 17

Integrated Pest Management follows which sequence?

A. Prevention and surveillance, threshold decision, multiple controls, and selective chemical use. B. Routine calendar spraying regardless of pest level. C. Fertiliser application followed by irrigation only. D. Complete rejection of biological control.

Answer: A.

Option-wise explanation:

- **A - Correct:** IPM combines compatible methods based on monitoring. The option states: Prevention and surveillance, threshold decision, multiple controls, and selective chemical use.
- **B - Incorrect:** Indiscriminate spraying can raise resistance. The option states: Routine calendar spraying regardless of pest level.
- **C - Incorrect:** Nutrition and irrigation are different. The option states: Fertiliser application followed by irrigation only.
- **D - Incorrect:** Biological control is a core option. The option states: Complete rejection of biological control.

Examiner trap 17: IPM is not zero control or maximum pesticide.

MCQ 18

Why are Custom Hiring Centres useful?

A. They transfer land ownership to the operator. B. They let small farmers rent machinery services and spread fixed cost across users. C. They guarantee machinery during every peak day. D. They are crop-insurance companies.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** Rental does not change title. The option states: They transfer land ownership to the operator.
- **B - Correct:** Service access can be more efficient than individual ownership. The option states: They let small farmers rent machinery services and spread fixed cost across users.
- **C - Incorrect:** Scheduling remains a constraint. The option states: They guarantee machinery during every peak day.
- **D - Incorrect:** Insurance has a separate function. The option states: They are crop-insurance companies.

Examiner trap 18: Sanctioned machinery is not timely service.

MCQ 19

Which is an institutional agricultural-credit source?

A. A tied village moneylender. B. An input trader's informal advance. C. A Regional Rural Bank. D. A landlord's consumption loan.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Moneylenders are non-institutional. The option states: A tied village moneylender.
- **B - Incorrect:** Trader credit is informal. The option states: An input trader's informal advance.
- **C - Correct:** RRBs are regulated banking institutions serving rural priority sectors. The option states: A Regional Rural Bank.
- **D - Incorrect:** Landlord lending is non-institutional. The option states: A landlord's consumption loan.

Examiner trap 19: Institutional status does not prove equal access.

MCQ 20

What is NABARD's principal system role?

A. Universal retail issue of every KCC. B. Fixing the repo rate. C. Operating PMFBY claims for every State. D. Refinance, rural development support and specified supervision within agricultural and rural finance.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** Banks and cooperatives issue KCCs. The option states: Universal retail issue of every KCC.
- **B - Incorrect:** RBI conducts monetary policy. The option states: Fixing the repo rate.
- **C - Incorrect:** Insurers and governments implement PMFBY. The option states: Operating PMFBY claims for every State.
- **D - Correct:** NABARD is the apex rural development financial institution. The option states: Refinance, rural development support and specified supervision within agricultural and rural finance.

Examiner trap 20: NABARD is not RBI or every farmer's branch bank.

MCQ 21

Under RBI's 2025 PSL Directions updated 7 August 2026, the agriculture target for applicable banks is:

A. 18% of ANBC or CEOBSE, whichever is higher, with nested NCF and SMF sub-targets. B. 18% of deposits with no denominator qualification. C. 10% for all agriculture and 18% for SMFs. D. A voluntary target without bank-category rules.

Answer: A.

Option-wise explanation:

- **A - Correct:** The official denominator and nested design matter. The option states: 18% of ANBC or CEOBSE, whichever is higher, with nested NCF and SMF sub-targets.
- **B - Incorrect:** PSL uses ANBC/CEOBSE. The option states: 18% of deposits with no denominator qualification.
- **C - Incorrect:** SMF is a sub-target, not the total. The option states: 10% for all agriculture and 18% for SMFs.
- **D - Incorrect:** The Directions are binding for covered banks. The option states: A voluntary target without bank-category rules.

Examiner trap 21: Do not detach a percentage from its denominator.

MCQ 22

Which statement correctly distinguishes KCC and MISS in 2025-26?

A. Every Rs 5 lakh KCC loan automatically carried 4% interest. B. Budget announced a Rs 5 lakh overall KCC limit, while the dated MISS concessional slab remained up to Rs 3 lakh. C. KCC is an insurance premium card. D. MISS abolished prompt-repayment conditions.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** The 4% rate requires eligible prompt repayment. The option states: Every Rs 5 lakh KCC loan automatically carried 4% interest.
- **B - Correct:** The product ceiling and interest-supported slab are different. The option states: Budget announced a Rs 5 lakh overall KCC limit, while the dated MISS concessional slab remained up to Rs 3 lakh.
- **C - Incorrect:** KCC is credit. The option states: KCC is an insurance premium card.
- **D - Incorrect:** PRI remains conditional. The option states: MISS abolished prompt-repayment conditions.

Examiner trap 22: Quote the ceiling, subsidised slab and year separately.

MCQ 23

A Joint Liability Group can improve farm credit access by:

A. Converting informal tenants into landowners. B. Waiving all appraisal and repayment. C. Using group responsibility and information where individual collateral or title is weak. D. Replacing banks with crop insurers.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Credit does not transfer ownership. The option states: Converting informal tenants into landowners.
- **B - Incorrect:** Banks still appraise loans. The option states: Waiving all appraisal and repayment.
- **C - Correct:** Group design can reduce information and collateral barriers. The option states: Using group responsibility and information where individual collateral or title is weak.
- **D - Incorrect:** Insurance and lending differ. The option states: Replacing banks with crop insurers.

Examiner trap 23: Group inclusion still needs credible cultivation evidence.

MCQ 24

Basis risk arises when:

A. The premium is paid before sowing. B. Every individual field is assessed perfectly. C. The insurer pays exactly the market price loss. D. An insured farmer's actual loss differs from the area-yield or weather-index result.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** Premium timing is not the definition. The option states: The premium is paid before sowing.
- **B - Incorrect:** Perfect assessment removes this mismatch. The option states: Every individual field is assessed perfectly.
- **C - Incorrect:** Crop insurance does not generally insure price. The option states: The insurer pays exactly the market price loss.
- **D - Correct:** Index and personal outcomes can diverge. The option states: An insured farmer's actual loss differs from the area-yield or weather-index result.

Examiner trap 24: Low verification cost trades off against basis risk.

MCQ 25

Which PMFBY premium pair is correct under official 2026 terms?

A. Kharif 2%; Rabi food/oilseed 1.5%; annual commercial/horticultural 5% maximum farmer shares. B. Kharif 5%; Rabi 5%; horticulture 1.5%. C. One uniform 2% rate for every crop. D. These percentages are claim-payment rates.

Answer: A.

Option-wise explanation:

- **A - Correct:** The seasonal/crop categories define farmer premium caps. The option states: Kharif 2%; Rabi food/oilseed 1.5%; annual commercial/horticultural 5% maximum farmer shares.
- **B - Incorrect:** The categories are reversed. The option states: Kharif 5%; Rabi 5%; horticulture 1.5%.
- **C - Incorrect:** Rates are not uniform. The option states: One uniform 2% rate for every crop.
- **D - Incorrect:** Premium share and indemnity differ. The option states: These percentages are claim-payment rates.

Examiner trap 25: Premium percentage is not claim percentage.

MCQ 26

PMFBY has been voluntary for farmers since:

A. Its 2016 launch only for non-loanee farmers. B. Kharif 2020. C. The 2021 farm-law repeal. D. The 2025 Seeds Bill consultation.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** The launch design differed. The option states: Its 2016 launch only for non-loanee farmers.
- **B - Correct:** The 2020 revamp made enrolment voluntary for loanee and non-loanee farmers. The option states: Kharif 2020.
- **C - Incorrect:** Farm-law repeal is unrelated. The option states: The 2021 farm-law repeal.
- **D - Incorrect:** Seed legislation is unrelated. The option states: The 2025 Seeds Bill consultation.

Examiner trap 26: Voluntary enrolment does not mean every State implements it.

MCQ 27

Conservation agriculture is built around:

A. Only maximum ploughing. B. Only organic certification. C. Minimum soil disturbance, permanent soil cover and crop diversification or rotation. D. Only drip irrigation.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Intensive tillage contradicts the first principle. The option states: Only maximum ploughing.
- **B - Incorrect:** Certification is a different system. The option states: Only organic certification.
- **C - Correct:** The three principles make it a cropping-system approach. The option states: Minimum soil disturbance, permanent soil cover and crop diversification or rotation.
- **D - Incorrect:** Water technology alone is insufficient. The option states: Only drip irrigation.

Examiner trap 27: Zero tillage is one practice, not the whole system.

MCQ 28

Which statement correctly distinguishes organic and natural farming?

A. The two labels are legally identical everywhere. B. Natural farming always permits every synthetic pesticide. C. Organic farming means no nutrient management. D. Organic follows defined standards/certification; natural farming is a distinct low-external-input approach and may not be certified organic.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** They cannot be universally merged. The option states: The two labels are legally identical everywhere.
- **B - Incorrect:** Natural farming seeks reduced synthetic inputs. The option states: Natural farming always permits every synthetic pesticide.
- **C - Incorrect:** Organic systems still manage nutrients. The option states: Organic farming means no nutrient management.
- **D - Correct:** Different standards, verification and input philosophies require separation. The option states: Organic follows defined standards/certification; natural farming is a distinct low-external-input approach and may not be certified organic.

Examiner trap 28: Low chemical use is not automatically certified organic.

MCQ 29

Rainfed Area Development under NMSA emphasises:

A. Integrated Farming Systems combining crops with suitable allied enterprises. B. Only major canal construction. C. Mandatory monoculture. D. Commodity futures trading.

Answer: A.

Option-wise explanation:

- **A - Correct:** IFS diversifies production and risk in rainfed settings. The option states: Integrated Farming Systems combining crops with suitable allied enterprises.
- **B - Incorrect:** Canal projects belong elsewhere. The option states: Only major canal construction.
- **C - Incorrect:** Diversity is central. The option states: Mandatory monoculture.
- **D - Incorrect:** Financial derivatives are unrelated. The option states: Commodity futures trading.

Examiner trap 29: IFS is not merely crop rotation.

MCQ 30

Climate-smart agriculture seeks:

A. One universal technology package. B. Productivity, adaptation and mitigation benefits where feasible and context-appropriate. C. Mitigation only, regardless of livelihood. D. A guarantee against every climate loss.

Answer: B.

Option-wise explanation:

- **A - Incorrect:** Climate risks and systems vary. The option states: One universal technology package.
- **B - Correct:** The framework balances three objectives with local trade-offs. The option states: Productivity, adaptation and mitigation benefits where feasible and context-appropriate.
- **C - Incorrect:** Adaptation and productivity matter. The option states: Mitigation only, regardless of livelihood.
- **D - Incorrect:** No technology guarantees all loss prevention. The option states: A guarantee against every climate loss.

Examiner trap 30: CCAFS was a CGIAR research programme, not an Indian scheme.

MCQ 31

Which is the best risk-layering approach?

A. Insure every predictable small cost. B. Use loan waivers as the only ex ante tool. C. Use prevention and savings for frequent loss, insurance for suitable uncertain loss, and public relief for catastrophe. D. Replace irrigation with insurance.

Answer: C.

Option-wise explanation:

- **A - Incorrect:** Insurance is poor for routine cost. The option states: Insure every predictable small cost.
- **B - Incorrect:** Waivers are retrospective and distortive. The option states: Use loan waivers as the only ex ante tool.
- **C - Correct:** Different frequencies and severities need different instruments. The option states: Use prevention and savings for frequent loss, insurance for suitable uncertain loss, and public relief for catastrophe.
- **D - Incorrect:** Risk reduction and transfer complement each other. The option states: Replace irrigation with insurance.

Examiner trap 31: Insurance cannot carry every farm-policy objective.

MCQ 32

A balanced input-subsidy reform should:

A. Remove all support overnight without alternatives. B. Ignore groundwater and nutrient effects. C. Treat fertiliser, power, credit and insurance subsidies as identical. D. Protect vulnerable farmers while correcting relative-price distortions and resource externalities.

Answer: D.

Option-wise explanation:

- **A - Incorrect:** Abrupt withdrawal can damage livelihoods. The option states: Remove all support overnight without alternatives.
- **B - Incorrect:** Behavioural externalities matter. The option states: Ignore groundwater and nutrient effects.
- **C - Incorrect:** Subsidies have different channels and incidence. The option states: Treat fertiliser, power, credit and insurance subsidies as identical.
- **D - Correct:** Transition design must balance equity, productivity and sustainability. The option states: Protect vulnerable farmers while correcting relative-price distortions and resource externalities.

Examiner trap 32: Compensation and sequencing are part of reform.

PYQS AND ANSWER PRACTICE

VERIFIED ROUTED MAINS DEMANDS

These demands are verified through audited official-paper routing ledgers. The formulations are neutral demand renderings and the solutions are original models, not official UPSC answers.

2018 GS-III ROUTE - SIKKIM ORGANIC TRANSITION

Demand: Discuss ecological and economic benefits of Sikkim's organic-State experience.

Model solution: Sikkim's transition followed a 2003 Assembly resolution, phased withdrawal of chemical-input support, extension and certification before the January 2016 full-organic declaration. Potential ecological gains include reduced synthetic residues, soil-biological recovery and less water contamination; economic gains include branding, tourism and premium-market opportunity. Yet transition yields, labour, certification cost and market access matter. Sikkim's small cultivated area, hill ecology and administrative capacity limit mechanical replication. The lesson is phased, certified and market-linked transition rather than an abrupt national ban.

2019 GS-III ROUTE - INTEGRATED FARMING SYSTEMS

Demand: Explain how Integrated Farming Systems can sustain agricultural production.

Model solution: IFS combines crops with suitable livestock, horticulture, fisheries or agroforestry so outputs and residues circulate across enterprises. Diversified products spread weather and price risk, improve year-round cash flow and use family labour. Manure and crop residues can support soil fertility, while farm ponds or trees add resilience. However, biological cycles, skill, veterinary support, markets and working capital make integration management-intensive. It should therefore follow local agro-climate and household resources rather than a standard enterprise bundle.

2019 GS-III ROUTE - WATERSHED DEVELOPMENT

Demand: Elaborate the impact of watershed development on water-stressed agriculture.

Model solution: Watershed development treats a common catchment from ridge to valley through vegetation, bunding, drainage-line treatment, recharge and livelihood measures. It can reduce erosion, improve soil moisture, raise protective irrigation and stabilise rainfed production. Community institutions can coordinate shared assets and grazing or water rules. Outcomes weaken where structures are scattered, maintenance lapses, tenure is unequal or downstream effects are ignored. Impact should be measured through hydrology, crop resilience and equitable access, not the number of structures alone.

2020 GS-III ROUTE - JAL SHAKTI ABHIYAN

Demand: State the principal features of Jal Shakti Abhiyan for water conservation.

Model solution: Jal Shakti Abhiyan began in 2019 as a campaign in identified water-stressed districts around rainwater harvesting, renovation of traditional water bodies, reuse and recharge, watershed development and afforestation. From 2021, Catch the Rain expanded mobilisation across districts with mapping, desilting and local water-body work. Campaign visibility can coordinate departments and communities, but durable water security depends on scientific siting, aquifer budgets, maintenance and crop-water incentives. Structures and geo-tags are intermediate outputs, not proof of recharge.

2020 GS-III ROUTE - WATER STORAGE AND IRRIGATION

Demand: Suggest measures to improve water storage and irrigation under depletion.

Model solution: Policy should combine catchment treatment, tank restoration, reservoir and canal maintenance, field channels, drainage, aquifer recharge and demand management. Micro-irrigation and tensiometer-based scheduling can improve application, while metering, feeder reform and crop incentives address rebound and over-pumping. Water Users Associations can improve distribution if authority and finance are real. Groundwater plans need recharge-extraction budgets at aquifer scale. Storage creation must be paired with silt control, maintenance and equitable access.

2021 GS-III ROUTE - MICRO-IRRIGATION

Demand: Assess how far micro-irrigation can address India's water crisis.

Model solution: Drip and sprinkler systems can reduce non-beneficial loss, improve application control and enable fertigation, especially for suitable crops and terrain. They can raise water productivity and reduce labour. However, capital cost, clogging, energy, maintenance and fragmented service constrain adoption. More importantly, lower water use per hectare may induce area expansion or water-intensive cropping, so aquifer withdrawal need not fall. Micro-irrigation must be combined with water budgets, measurement, crop alignment and community groundwater governance.

2022 GS-III ROUTE - IFS FOR SMALL FARMERS

Demand: Explain the benefits of Integrated Farming Systems for small and marginal farmers.

Model solution: Small farms can combine crops with dairy, poultry, horticulture, fisheries or agroforestry to diversify income, recycle nutrients and use labour across seasons. Multiple outputs reduce dependence on one harvest and can improve household nutrition. Enterprise complementarities may lower purchased-input cost. Yet smallholders face working-capital, skill, animal-health, storage and market constraints. IFS succeeds when the enterprise mix matches land, water, labour and demand; simply adding activities can increase complexity and risk.

2024 GS-III ROUTE - IRRIGATION CHALLENGES

Demand: Examine challenges of India's irrigation system and government measures.

Model solution: India's irrigation challenge includes incomplete projects, gaps between potential created and utilised, canal maintenance, head-tail inequity, waterlogging, fragmented minor systems and groundwater over-extraction. PMKSY links AIBP, HKKP and watershed work, while PDMC under RKVY supports micro-irrigation. CADWM, tank restoration, user associations and aquifer management address different levels. Reform must measure reliable field service and water productivity, correct power and crop incentives and protect small farmers during pricing transitions.

2025 GS-III ROUTE - GROUNDWATER DEPLETION

Demand: Examine causes of groundwater depletion and government responses.

Model solution: Groundwater depletion reflects individually flexible pumping from a shared aquifer, cheap or unmetered power, water-intensive crop incentives, uncertain rainfall, weak recharge and limited measurement. Borewell deepening then raises energy cost and excludes poorer farmers. Atal Bhujal Yojana, aquifer mapping, watershed recharge, micro-irrigation and community water budgets are relevant responses. Yet recharge structures cannot offset unlimited extraction. Sustainable policy must align power, procurement, crop choice, measurement and local withdrawal rules.

OBJECTIVE PYQ ROUTING - KEY WITHHELD WHERE NOT OFFICIALLY VERIFIED

Year / paper	Routed official-paper demand	Key discipline
2018 Prelims GS-I Q59	Conservation Agriculture practices	Answer withheld pending official UPSC key.
2019 Prelims GS-I Q39	Carbofuran, phorate and triazophos classification	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q59	District Central Cooperative Banks and agricultural credit	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q66	Kisan Credit Card eligible short-term purposes	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q80	Biochar uses in farming	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q83	Zero tillage benefits and limits	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q89	Sugarcane bud-chip and tissue-culture propagation	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q90	Crop diversification and tensiometer	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q91	Fertigation	Answer withheld pending official UPSC key.
2020 Prelims GS-I Q94	Fertiliser chemistry and pricing	Answer withheld pending official UPSC key.
2021 Prelims GS-I Q51	Permaculture distinction	Answer withheld pending official UPSC key.
2021 Prelims GS-I Q59	Climate-Smart Agriculture, CCAFS and ICRISAT	Answer withheld pending official UPSC key.
2026 Prelims GS-I Q28	Rainfed Area Development and Integrated Farming Systems	Answer withheld pending official UPSC key.

ORIGINAL MAINS PRACTICE WITH MODEL SOLUTIONS

ORIGINAL MAINS 1 - 10 MARKS

Question: Why can micro-irrigation improve field efficiency without reducing groundwater depletion? Answer in 150 words.

Model answer:

Micro-irrigation applies water more precisely through drip or sprinkler systems, reducing uncontrolled conveyance, evaporation and application losses at farm level. Per Drop More Crop supports this technology under RKVY from 2022-23.

Yet aquifer depletion depends on total extraction, not litres applied per hectare. Lower unit water cost can encourage farmers to expand irrigated area, grow more seasons or retain water-intensive crops. Subsidised electricity and weak pumping measurement reinforce this rebound. Some apparent field losses also return to the aquifer or downstream system, so gross application saving is not identical to basin saving.

Micro-irrigation should therefore be combined with aquifer budgets, crop incentives, reliable power measurement, water-user institutions and extension. The correct metric is sustainable consumptive use and groundwater level, not equipment installed.

Native-body word count: 142 / 150.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish KCC, priority-sector lending and interest subvention. Answer in 150 words.

Model answer:

Kisan Credit Card is a revolving bank-credit product for crop and eligible allied working-capital needs. Priority-sector lending is an RBI portfolio obligation: under the 2025 Directions updated 7 August 2026, applicable banks face an 18 per cent agriculture target with nested non-corporate and small-farmer sub-targets.

Modified Interest Subvention Scheme changes the price of eligible short-term KCC credit. For FY2025-26, the concessional structure retained 7 per cent interest up to Rs 3 lakh and a 3 per cent prompt-repayment incentive, reducing the effective rate to 4 per cent for eligible timely payers. Budget 2025-26 separately announced a Rs 5 lakh overall KCC limit.

Thus product access, bank target and interest support are distinct. None guarantees sanction, drawal or inclusion of undocumented tenants.

Native-body word count: 141 / 150.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.

ORIGINAL MAINS 3 - 15 MARKS

Question: Evaluate PMKSY as an irrigation and water-management framework. Answer in 250 words.

Model answer:

PMKSY was designed as a convergence framework linking project completion, irrigation access, command management, watershed treatment and on-farm efficiency. Its 2021-26 architecture included AIBP, Har Khet Ko Pani and the Watershed Development Component. Per Drop More Crop, originally associated with PMKSY, has operated under RKVY from 2022-23.

AIBP addresses delayed major and medium projects. HKKP includes command-area development, surface minor irrigation and repair or restoration of water bodies, while WDC treats rainfed catchments from ridge to valley. These components correctly recognise that a dam, a field channel, a tank, a watershed and a drip system solve different constraints.

The limitation is the gap between potential created and reliable service. Canal systems face siltation, head-tail inequity, waterlogging and weak maintenance. Micro-irrigation can produce rebound if irrigated area expands. Watershed structures fail without community ownership, and groundwater works are inappropriate in stressed aquifers.

Official June 2026 material continues to use the PMKSY umbrella after the 2021-26 approval window, but component-specific funding and institutional location must be dated. Reform should measure utilised potential, tail-end delivery, water productivity, aquifer balance and maintenance rather than expenditure or hectares alone. PMKSY succeeds when infrastructure, incentives and local water governance converge.

Native-body word count: 214 / 250.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse India's fertiliser-subsidy architecture and nutrient imbalance. Answer in 250 words.

Model answer:

India uses different subsidy and pricing regimes. Urea has a government-controlled maximum retail price and a separate subsidy mechanism. Eligible phosphatic and potassic fertilisers operate under Nutrient Based Subsidy, where periodic per-kilogram rates are notified for nitrogen, phosphorus, potassium and sulphur while product MRPs are decontrolled subject to oversight.

This asymmetry makes nitrogen relatively cheap and can encourage excessive urea use compared with phosphorus, potassium, sulphur, micronutrients and organic matter. The result may be lower marginal response, soil imbalance, nitrate pollution and fiscal exposure. However, a national N:P:K ratio cannot prescribe every soil or crop.

Fertiliser DBT improves transaction tracking but is not cash transferred to farmers: companies receive subsidy after authenticated retail sale through PoS, while farmers buy at subsidised prices. Soil Health Cards and integrated nutrient management can link mineral, organic and biological sources to measured deficiency. Split placement, fertigation and extension improve use efficiency.

Reform should gradually correct relative prices while protecting small farmers, strengthen soil testing and quality control, and redirect savings toward research and balanced nutrients. Nano urea should follow crop-soil recommendations; notification or sale does not prove universal replacement. The objective is nutrient productivity and soil health, not maximum fertiliser volume.

Native-body word count: 219 / 250.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.

ORIGINAL MAINS 5 - 20 MARKS

Question: Assess PMFBY's design, achievements and limitations as crop-risk protection. Answer in 250 words.

Model answer:

PMFBY pools specified crop-production risks for notified crops, areas and seasons. Official 2026 terms retain maximum farmer premium shares of 2 per cent for Kharif, 1.5 per cent for Rabi food and oilseed crops and 5 per cent for annual commercial or horticultural crops; government subsidises the balance. Enrolment has been voluntary for farmers since Kharif 2020.

For widespread loss, the scheme primarily uses area-yield assessment based on Crop Cutting Experiments. This lowers the cost of verifying millions of individual farms and limits moral hazard. Specified prevented-sowing, localised and post-harvest losses follow separate reporting and assessment routes. YES-TECH, WINDS and CROPIC seek better yield, weather and field evidence.

The central weakness is basis risk: an individual may lose crop while the insurance-unit yield does not cross the trigger. Delayed CCEs, State premium release, notification gaps and inaccessible grievance can further weaken trust. Voluntary State and farmer participation can also narrow the pool.

Reform requires smaller credible units where feasible, independent data audits, timely premium funding, transparent claim calculation and accessible appeals. Insurance should be layered with irrigation, resilient seed, diversification, savings and catastrophe relief. PMFBY protects covered production risk; it is not price insurance or a universal income guarantee.

Native-body word count: 220 / 250.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an integrated strategy for productive, equitable and sustainable agriculture. Answer in 250 words.

Model answer:

An integrated strategy must align water, seed, nutrients, machinery, credit, insurance, extension and markets rather than maximise one input. Water policy should complete viable projects, modernise command areas, restore tanks and watersheds and govern aquifers. Micro-irrigation needs maintenance and extraction rules so field efficiency becomes basin saving.

Seed systems should combine certified quality with affordable farmer-saved and locally adapted material. Fertiliser reform should correct the urea-P&K price imbalance, expand soil testing and integrated nutrient management and treat nano products through independent crop-soil evidence. IPM should use surveillance and thresholds before selective chemicals. Custom Hiring Centres can make machinery accessible without inefficient individual ownership.

Credit architecture should join RBI priority targets, KCC, cooperatives, RRBs, commercial banks, NABARD refinance and JLG or SHG inclusion. The overall KCC ceiling, concessional MISS slab, sanction and disbursement must remain distinct. PMFBY should reduce basis risk and settlement delay while remaining one layer of risk management.

Conservation agriculture, agroforestry, Integrated Farming Systems, millets and climate-resilient research should be selected by agro-climate and markets. Natural and organic farming require phased transition and nutrient evidence. Input-subsidy reform must protect vulnerable farmers while correcting groundwater, nutrient and fiscal externalities. Cropping details remain Topic 11 and digital-governance architecture Topic 27.

Native-body word count: 221 / 250.

Answer audit: Definition and thesis -> named statutory or institutional evidence -> economic analysis -> qualification -> concise conclusion.